

General Industries Hub

Enterprise Inventory Management

HOD Manual

Head of Department oversight: approvals, EOD, reports.

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Confidential - for authorized personnel only

GI Hub - Role booklet - HOD Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
1	Quality Control (QC)	One site or one warehouse - never both
2	Head of Department	One site
2	Head of Qualities (QC-HOD)	All sites, Surface Shield only, read + escalate
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

The Head of Qualities is level 2 but reads across every site, and that is not a contradiction. Cross-site reach normally comes from being level 3 - and level 3 would also have handed this role every endpoint the rank admits, which is most of the system. Instead it is level 2 with a named cross-site exemption, and the rank grants it nothing at all: a level check refuses it outright, so it reaches only what names the role explicitly. Its whole surface is the Quality Oversight page. See §23.

The QC role is scoped on one of two axes, and exactly one. Material is inspected either where it arrives (a warehouse) or where it lands (a site), and which one depends on how your operation is organised. A QC account is therefore created against a site or against a warehouse, and an account carrying neither - or both - sees nothing at all rather than everything. That is deliberate: an inspector with an unclear remit is a bigger problem than an inspector with none.

2.2 Page access matrix - which role can open which page

[!] A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Read that caveat together with the table below it, not instead of it: the caveat says a page is locked to "its own role", and the table is what names that role. A Head of Department can open Man-Hours and the Material Estimator; those two are locked to the HOD, not away from them. Admin is the single exception and reaches every workspace deliberately, for support.

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Live Dashboard	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Stock	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[X]	[OK]	[OK]
Lots & Expiry (\$3.10.3)	[OK]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[X]	[OK]
Rack Locator	[OK]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Assets	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Entry Log (all tabs)	[OK]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
HOD Portal -> Approvals	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
HOD Portal -> Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Lining Coverage	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Bulk	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Excel Import									
WBS & Work Types	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Material Estimator (SME)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Man-Hours / Manpower Tracking	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Execution Entries	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[X]	[OK]
Reports	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK] (read + download)	[OK]
Logistics Portal (Procurement)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Warehouse Portal (Receiving & DN)	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Quality -> Inspections	[OK] (read)	[OK] (read)	[X]	[OK] (decides)	[OK] (read)	[OK] (read)	[X]	[OK] (read)	[OK]
Quality -> QC Accounts	[X]	[OK]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Quality Oversight (\$23)	[X]	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK]
Safety & People -> PPE Forecast	[OK]	[X]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]
Safety & People -> PPE Usable Time	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Safety & People -> Employees	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Records -> Inventory	[OK]	[OK]	[OK]	[OK]	[OK] (+ New item / Edit, own site)	[OK]	[OK]	[OK] (read)	[OK]
Records -> Receipts, Consumption, Returns, Lots, Purchase Requests	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK] (read)	[OK]
Records -> Purchases	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[OK] (read)	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
e Orders									
Valuation & 30-Day Burn (\$24.3)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Admin Portal	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Documents - Security - Training - Feedback	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]

Where each role lands after signing in: Store Keeper -> Entry Log -> Issue Stock - Warehouse User -> Warehouse Portal - Supervisor -> Supervisor Portal - QC -> Quality -> Inspections - HOD -> HOD Portal -> Approvals - Logistics -> Logistics Portal - Head of Qualities -> Quality Oversight - Auditor -> Dashboard - Admin -> Admin Portal -> Console.

2.3 What each role can do

One list per role, in plain words. If you only want to know whether your account can do a thing, read your own entry - each is complete on its own.

2.3.1 Store Keeper - what a Store Keeper can and cannot do

Can: open the Entry Log and post every stock movement - receive, issue, return, adjust, count, returnable items, OCR import; see Incoming Deliveries and Supervisor Requests; open the Dashboard, Stock, Rack Locator, Assets, PPE Forecast, PPE Usable Time, Employees and Execution Entries; open Lots & Expiry and Records -> Inventory; read quality inspections; download documents.

Cannot: approve anything (an HOD approves what a Store Keeper stages); open the HOD, Supervisor, Logistics or Warehouse portals; open Reports, the ledger Records (Receipts, Consumption, Returns, Lots), the Material Estimator or Man-Hours; see another site's stock.

Scope: one site.

2.3.2 Supervisor - what a Supervisor can and cannot do

Can: raise material requests from the Supervisor Portal; post Execution Entries; open the Dashboard, Stock, Assets and Employees.

Cannot: post stock movements (the Entry Log is the Store Keeper's); approve requests; open Reports, the ledger Records, Lots & Expiry, the Material Estimator, Man-Hours or any other role's portal.

Scope: one site.

2.3.3 Quality Control (QC) - what an inspector can and cannot do

Can: decide inspections - pass, fail or hold - from Quality -> Inspections; open the Stock page to see what is waiting and what their decision released; open Lots & Expiry to see each lot's expiry and what is left of it (§3.10.3).

Cannot: open the Dashboard, post any stock movement, approve a purchase, create QC accounts, or move themselves between sites (that is an Admin decision). See §22.

Scope: exactly one site or exactly one warehouse - never both, never neither.

2.3.4 Head of Department (HOD) - what an HOD can and cannot do

Can - and this is the question people ask most: open the Man-Hours / Manpower Tracking page and the Material Estimator. Both are exact-locked to the HOD (and Admin for support); the lock is what admits an HOD, not what excludes them. That includes all thirteen Man-Hours tabs - roster, timesheets, estimator, variance, scorecard, employee-wise, the four execution views, the Manpower Planner and the SME Session Plan (§19).

Also: approve or reject everything the site stages (receipts, issues, returns, adjustments, counts); create and submit Purchase Requisitions to Logistics; raise Cross-Site Requests; run Reports; open the Executive Summary, Burn Rate, Low Stock, Lining Coverage, Document Library and Bulk Excel Import; create QC accounts for their site; post Execution Entries; open PPE Forecast, PPE Usable Time and Employees; open Lots & Expiry (§3.10.3); add and edit inventory items for their own site from Records -> Inventory (deleting stays Admin-only).

Cannot: open the Logistics Portal, the Warehouse Portal, the Supervisor Portal, the Entry Log, Quality Oversight or the Admin Portal; raise a Purchase Order (Logistics does that); see another site's stock - though they may request material from one.

Scope: one site.

2.3.5 Head of Qualities (QC-HOD) - what this role can and cannot do

Can: open the Quality Oversight page and, since Phase 16, Lots & Expiry (§3.10.3) - every lot's expiry and what is left. Across every site and warehouse, but only for Surface Shield material: the controlled category is filtered in the database on every query, so it is the boundary of the role rather than a filter on a page. Seven views - overview, Surface Shield purchase orders, the MTC register, where it is used, stagnation and expiry, escalations, and the 90/60-day thresholds. They may raise and close escalations and retune those two thresholds.

Cannot: approve or reject a quality inspection; move, issue or receive stock; raise a PR or PO; open any other portal or page; see PPE, tools, consumables or any other category. See §23.

Scope: all sites and all warehouses; Surface Shield only.

2.3.6 Logistics Coordinator - what Logistics can and cannot do

Can: run the whole procurement chain from the Logistics Portal - receive submitted PRs, raise Purchase Orders, assign POs to warehouses, manage vendors, track DN status; open Records (inventory, ledgers, POs, PRs), Reports, the Warehouse Portal, Lining Coverage, the Rack Locator, PPE Forecast, and quality inspections; create QC accounts.

Cannot: open the HOD Portal, the Entry Log, the Supervisor Portal, the Material Estimator, Man-Hours or the Admin Portal - outranking a role does not open its workspace.

Scope: all sites, no site lock.

2.3.7 Warehouse Operator - what a Warehouse User can and cannot do

Can: receive against POs, prepare and dispatch Delivery Notes from the Warehouse Portal; open the Dashboard, Stock, Rack Locator and Assets; read quality inspections; create QC accounts for their warehouse.

Cannot: open the Logistics Portal, the HOD Portal, the Entry Log, Reports, Records, the Material Estimator or Man-Hours.

Scope: one warehouse.

2.3.8 Auditor - what a view-only Auditor can and cannot do

Can: read across every site - Dashboard, Stock, Assets, Records, Reports (including downloads), Lining Coverage, the Material Estimator, the HOD Portal's read-only views (Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site Requests), Employees and quality inspections.

Cannot: change anything at all, anywhere. Every write route is refused by the server, not merely hidden. Not Man-Hours, not the Entry Log, not any approval. See §20.

Scope: all sites, read-only.

2.3.9 Administrator - what an Admin can do

Everything, including every other role's workspace as a support shadow, plus the Admin Portal: users, access requests, overdue actions, inventory master, the audit log and the console.

Adding an inventory item (Admin -> Inventory Master -> New item). Pick the Site and the Category from the suggestions. The server matches them to the spelling already in use, ignoring capitals and a plural s, so cncec is saved as CNCEC and consumable as Consumables. This matters because a store keeper only sees items whose site matches theirs exactly. If you type a site or category that no item uses yet, GI Hub asks "Create a new site...?" first. Press Cancel if you meant an existing one. A new item must have a site.

The HOD adds and edits items too (30 September 2026). On the Inventory page (Records -> Inventory), the Admin and the HOD have a New item button and an Edit button on each row. The same form, the same Site / Category matching and the same audit log apply. The HOD's items are always saved to their

own site, which the form shows instead of asking. A HOD cannot add or edit another site's items. Deleting an item stays Admin-only (Admin -> Inventory), because it cannot be undone.

2.4 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Dashboard, Stock, Material Requests and Execution Entries are all site-locked.
Quality Control (QC)	Whichever single place they were created against: one site, or one warehouse. A site QC sees inspections raised at that site and has no warehouse business at all; a warehouse QC is pinned to its warehouse exactly as a Warehouse User is. An account with no binding, or with both, sees an empty list - it never falls back to seeing everything. Moving a QC to another site is a request, decided by an Admin, not something the QC or their HOD can do alone.
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab) and submit PRs to Logistics.
Logistics	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Head of Qualities	All sites globally, but ONLY Surface Shield material - every query on their page is filtered to the controlled category in the database, so the category is the boundary of the role rather than a filter on a page. They read, and they send escalations; they cannot approve an inspection, move stock, raise a PR or open any other portal. See §23.
Auditor	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It can open the Dashboard, Stock, Records, Reports and Lining Coverage and change nothing anywhere. See §20.
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar when shadowing the Warehouse Portal.

A site-locked user is never asked for their site (30 September 2026). On Issue, Receive, Return, Adjust, the supervisor's material request and the HOD's purchase request, a Store Keeper, Supervisor or HOD sees their site as a blue label (CNCEC your site), and the entry is saved there. The same goes for the

"All sites" filters on Approvals, Burn Rate, Reports, SME and the Executive Summary. Picking another site used to be possible and only produced an error, because the server allows your own site only. An account with no site assigned still sees the dropdown; ask an Admin to set the site.

2.5 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics, warehouse_user, qc, qc_hod or auditor accounts are seeded. All of them are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Live | Practice switch - above the username. Live is the real system and is selected unless you choose otherwise. Practice is a separate training copy with made-up data, where nothing you do affects Live. Switching signs you out. See §26 Practice Mode for everything about it.
- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open. This works from any address, including a bookmarked /login; an address that is not a page takes you to your home page instead of a blank screen.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.1.1 The look of the sign-in page

Phase 14e, redesigned in Phase 15c.

The large gold GI mark sits in its own space at the top of the page, and the glass sign-in card sits below it, so the two never overlap. The mark rises into place once when the page opens. After that it stands still, and a slow band of light sweeps across the gold every few seconds. Nothing on the page follows the mouse. What you see depends on your device, and the form works the same on all of them, from the first moment:

- Everyone, including the Android, iOS and desktop apps: the mark drawn in the page's style.
- A desktop computer with a graphics card: the same mark in 3D gold, in the same place. It loads separately after the page is ready, so it never delays signing in. It draws nothing between sweeps, stops when you switch tabs, and is removed as soon as you sign in.
- A very short window (under about 560 pixels tall): the large mark gives way to a small, still mark inside the card.
- Anyone with Reduce motion switched on in their system settings: no rise and no sweep. The mark simply stands there.

The Executive Summary and the Training page use the same glass style. Their cards rise gently into place as you scroll, except with Reduce motion on and when printing.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

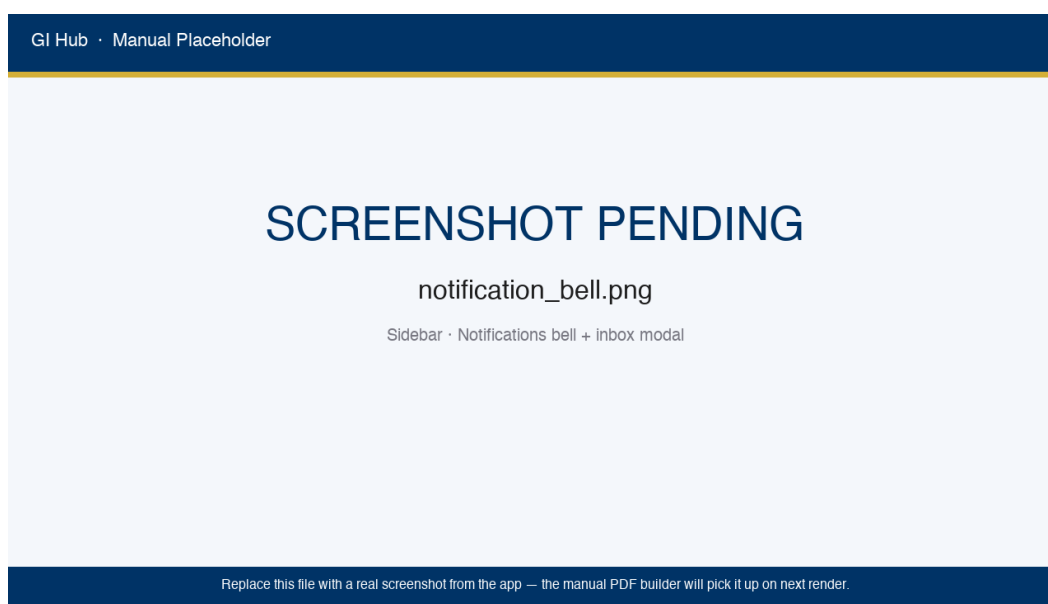
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

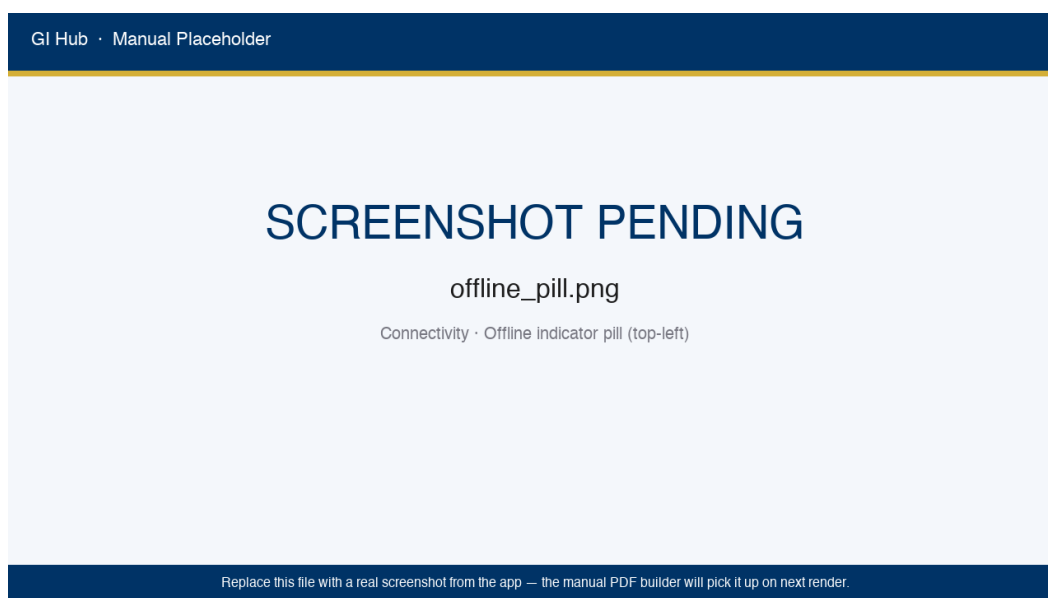
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line

Event you'll see	Triggered by
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

3.7 Saving an entry when the signal drops

If the network drops while you save a receipt, issue, return, stock adjustment or bulk batch, GI Hub keeps the entry on your device and shows a saved offline message instead of an error. A badge in the header shows how many entries are waiting. They are sent by themselves when the connection returns, and you can press the badge to send them straight away.

An entry is never saved twice. Each save carries its own reference, and the server recognises it if it arrives again. That can happen when the page reloads while the entry is being sent, or when the connection drops just as the server confirms. The second arrival is matched to the first, and only one entry reaches your HOD. You never need to check for, or delete, a duplicate caused by a bad connection.

3.8 What's new

Phase 14d.

When a feature you use changes, a What's new panel opens by itself the next time you sign in. It shows only what concerns you: an announcement is sent to the roles that can open the page it is about, so nobody is told about a screen they cannot reach.

Each card says what changed. Open it takes you to the page, Watch the tutorial plays the short video when there is one, and a User Manual § tag names the section to read.

Press Got it and the panel closes. It does not open by itself again for the same announcements. To see recent ones, press the gift icon in the top bar, next to the bell; a number on it means something new is waiting.

A matching notification also appears in the bell (§3.6). Announcements are in-app only - never WhatsApp or email.

3.9 Stock vs the Excel workbook - which materials differ, and why

Added 26 September 2026.

GI Hub keeps its own stock ledger, and the site also keeps CNCEC_Inventory.xlsx. After every Excel sync, GI Hub compares its stock for every material with the workbook's Current Stock column. It then works out why any material differs.

Where: the top of the Stock page.

- A yellow banner says how many materials differ, for example 14 of 505 materials don't match the Excel workbook, and when that was checked.
- Each of those materials has a red row and a red != Excel tag. Point at the tag to see the reason.
- Review the differences lists every material. Each one shows its workbook and GI Hub figures, what caused the difference, where to look (a GI Hub record number or an Excel sheet and row) and how to fix it.
- A material's own page (scan its QR or open it from Stock) shows the same warning.

What the causes mean:

Tag	What happened	How to fix it
Only in GI Hub	A receipt, issue or return was entered in GI Hub, and the workbook's log never got it. The entry names who made it and its remarks.	If it really happened, add it to the log. If it was a test or a mistake, reverse it in GI Hub with a Stock Adjustment, or have an admin remove it.
No longer in Excel	GI Hub took a line from the workbook, and the workbook has since lost it (deleted or re-dated).	Put the line back if that was a mistake. Otherwise an admin removes it with the sync's --prune-vanished option.
Not synced yet	The workbook has a line GI Hub doesn't have yet.	Run the Excel sync.
Edited in Excel	A quantity was changed in the workbook after the last sync.	Run the Excel sync.
Sheet totals stale	The Inventory sheet's own Current Stock doesn't match its own logs. Current Stock is a typed value in this workbook, so it can fall behind.	Correct the figure on the Inventory sheet.
Opening stock	The Opening Stock differs.	Run the sync, or correct the workbook.
Unexplained	Part of the difference that none of the above explains. It is always shown, never hidden.	Compare the material's records by hand.

The marked workbook. Press Get the marked workbook and choose your CNCEC_Inventory.xlsx. You get back a copy containing:

- a GI Hub check sheet first, listing every cause with how to fix it;
- each differing material's row on the Inventory sheet filled red, with a note on its Current Stock cell;
- any log line the check names filled amber.

Make the fixes in your own workbook, then sync again. Your file itself is never changed. Check again uploads the workbook and refreshes the list without downloading anything.

Store Keepers, Warehouse, HODs, Logistics and Admins can upload. Everyone who can see Stock can read the result. Nothing on this screen changes stock.

[!] The check only explains. It never corrects stock by itself, because deciding whether an entry was real is a person's call.

3.10 Lots, FEFO and the Lot Register workbook

Phase 16 (October 2026).

A lot is one batch of a material: the cans of PU COMP A from batch 3504, or the CHEMOLINE rolls of batch 1O25003382. GI Hub keeps the stock of each lot, so it can tell the store keeper which lot to use first (FEFO, first expire, first out).

3.10.1 What the `Serial No.` column means in the workbooks

The Receipt, Consumption and Return Logs have one Serial No. column. GI Hub reads it according to the item on that row:

The item is...	Serial No. is...	GI Hub keeps it as...
a Surface Shield (PU, Furan, Phenacin, Coroflake, BC 3004...)	the batch	the Lot No. (and the cell as typed)
a roll (CHEMOLINE, UOM ROL)	the roll number	the roll number, and its batch as the Lot No.
a brick (AR 30/40 MM, carbon bricks)	not a lot - bricks have no batch and never expire	the serial, with no lot
anything else (pumps, tools, equipment)	the equipment / asset no.	the serial, as before

Rules that keep this safe:

- A lot is spelled one way. 3504.0 is 3504, and A4525 and A 4525 are the same lot (A 4525). N/A or a blank cell means no lot.
- Roll numbers begin with 1O (the letter O). A roll typed 10... (with a zero) is corrected to 1O..., and the sync says so.
- One lot per cell. A cell such as 4525 = 55 Cans; 1823 = 2 Cans gets no lot. Split it into one row per lot in the workbook; the sync names the rows that need it.
- A lot that was never received is reported, not invented. If the Consumption Log names a lot no receipt brought in for that material, the sync lists it with the material it was received under (often a typing slip), and the consumption still counts toward stock.

- History keeps the lot it was given. GI Hub never picks a lot for an old workbook row that has none. Automatic FEFO applies only to issues entered in the app.
- A return gives back to its lot. The Return Log's Serial No. is now read too: a Surface Shield return carries its lot, and an equipment return keeps its asset tag.

Every receipt that names a lot creates that lot in GI Hub. A lot's stock is always received - consumed - returned, worked out from the ledger.

3.10.2 The Lot Register workbook ('Rubber & Brick Materials - CNCEC.xlsx')

This workbook keeps, per Surface Shield material, one sheet of what arrived: batch, manufacture date (MFD), expiry, DN No. and quantity. Put it in the same folder as the other workbooks. The normal sync reads it automatically; any file named like *Rubber*Brick*CNCEC*.xlsx is found, and the run prints a > lots section.

[!] It never changes stock. Stock comes only from the Receipt, Consumption and Return Logs. This workbook only describes the lots GI Hub already has: their MFD, expiry and batch. Its quantities are compared with the Receipt Log, and every difference is reported. A mistake in it can make an expiry wrong, but it can never double a quantity.

How each row is matched:

1. To its material - by the SAP column. If a sheet has no SAP column, GI Hub matches the material name, the component letter (COMP C) and the pack size (10 Kg vs 2.5 Kg), and refuses anything it cannot pin to exactly one item. The material code is only a cross-check, because COMP A-D share one code.
2. To its receipt - the same SAP, DN No. and received date. CHEMOLINE rolls have no DN, so they match on SAP and date, roll by roll.
3. To its lot - the Batch No. (normalised as in §3.10.1). For CHEMOLINE the Order No. is the batch and each Roll No. is a roll inside it.

What it fills in:

On...	It writes...
the lot	MFD, expiry, batch reference, DN No.
the roll register	each CHEMOLINE roll and the batch it belongs to
a receipt with no lot	the lot, when the workbook names exactly one batch for it
an item with no shelf life	the shelf life (months), learned from its own MFD -> expiry pairs

Expiry:

- An expiry in the workbook is used as it is (from the file).
- A missing expiry is MFD + the item's shelf life (derived).
- If two rows of one lot disagree, the earliest expiry is used.
- An expiry before its MFD is refused and reported.
- An expiry a person typed on the Receive form is never overwritten.

Bricks are compared with the receipts but get no lot (they have no batch and never expire).

What the sync tells you (each is listed by sheet and row):

- quantities that differ from the Receipt Log;
- rows with no matching receipt;
- expiries before the MFD;
- lots whose rows disagree on dates;
- material codes that disagree with the inventory;
- a batch the Receipt Log spells differently (e.g. 0926 vs 0.926);
- rows with no batch.

3.10.3 Lots & Expiry - which lot to use, and what is about to expire

The Lots & Expiry page (sidebar, under Stock) is for the Store Keeper, QC, the HOD and the Head of Qualities. It lists every lot of a lot-tracked material, the oldest expiry first, with:

- MFD and expiry - an expiry marked derived is MFD + the item's shelf life;
- status - Expired, <= 30 days, <= 60 days, <= 90 days, OK, No expiry, or Quarantined / Disposed when the Admin has taken the lot out of use (Admin Console -> Lots). A quarantined or disposed lot is never offered on the Issue form;
- received, consumed, returned and remaining - worked out from the ledger.

Click a status at the top to show only those lots. Show used-up lots adds the lots with nothing left. A CHEMOLINE batch opens to show its rolls: when each was received, where it is, and for a used roll the day and tank it was issued to. A red banner warns when an expired lot still has stock. At the bottom, Lots used but never received lists lots the Consumption or Return Log names that no receipt brought in.

Issuing (the Issue form's Lot field). For a lot-tracked material, the Lot field is a list of that material's open lots in FEFO order. Each shows its expiry, the days left and the quantity left.

- The first one is marked FEFO. Leaving the field blank issues from it, because GI Hub's own automatic pick uses the same order.
- [!] An expired lot is listed last, marked red, and is never the suggestion, even though its date is the earliest. You can still choose it; FEFO warns, it never blocks.
- Choosing any lot other than the FEFO lot asks for a reason, which goes to the HOD with the issue. Choosing the FEFO lot asks nothing.
- CHEMOLINE: pick the roll; its batch fills in by itself. A roll already issued is no longer offered.

Receiving (the Receive form). For a lot-tracked material the form asks for the batch (as printed on the label), the manufacture date (MFD) and the expiry:

- If you leave the expiry blank, it becomes MFD + the item's shelf life, and the form shows the date it will use.
- An expiry before the MFD, or an MFD in the future, is refused.
- An expiry you type is kept, and the Lot Register workbook never overwrites it.

The daily expiry notice. Each evening the Store Keeper and the HOD of a site get one notice listing that site's lots that expire within 30 days or have already expired, each with how much is left. It is one notice per site, never one per lot. On WhatsApp it arrives inside the evening summary.

Shelf life and lot tracking (item editor). Admin -> Inventory or Records -> Inventory -> Edit has two

settings:

- Lot tracking: Automatic - every Surface Shield except bricks - or Tracked by lot, or Not tracked.
- Shelf life (months): the number a missing expiry is derived from. The Lot Register workbook fills it in where it is empty, from the item's own MFD -> expiry pairs (PU 9, Phenacin 6, BC 3004 24...).

Reading the two warnings - they are different things.

You see...	It means...	What to do
a red Expired lot with stock left	The lot was received, its expiry has passed, and the ledger says some of it is still on the shelf.	Check the shelf. If it is there, ask the Admin to dispose of it or quarantine it. If it is not, an issue or return was never entered - enter it, or correct the workbook.
a lot under Lots used but never received	The Consumption or Return Log names a lot that no receipt of that material brought in. The lot itself may be fine; usually the lot number or the SAP code on that line was typed wrong. The column Received under SAP shows the material that did receive a lot with that number.	Correct the line in the workbook (the lot, or the SAP code) and sync again. The stock already counts the line either way.

An expired lot whose stock is all used shows Used up, not Expired, and appears only with Show used-up lots switched on. So when a lot shows received 1, consumed 1, remaining 0, there is nothing to dispose of.

3.10.4 Practising with lots

Practice (\$26) has lots to try all of this on:

- PRACTICE PU PRIMER (899971) has three lots: PR-OLD (expired, still has stock), PR-SOON (expires within 30 days, so it is the FEFO suggestion) and PR-LATE.
- PRACTICE CHEMOLINE (899973) has batch 1O26009999 with three rolls to pick.
- One issue names lot PR-TYPO, which no receipt brought in, so the Lots & Expiry page has a used but never received example.

3.11 What changed in September-October 2026 (Phases 14, 15 and 16)

A one-page list for everyone who used GI Hub before September 2026. Each line names who it affects and where it is explained in full.

3.11.1 Phase 14 - quantities, reconciliation and announcements (26 September)

Change	Who	Read
Surface Shield quantities are shown both ways, kilograms first (189 KG - 21 Can). You still enter packs; a consumption form may be written in KG and is converted.	Store Keeper, Supervisor, HOD	§4.11, §4.10.8
When a QR form and the Excel workbook both record the same drum, the ledger keeps one quantity (the larger), never both.	HOD	§6.17
The Surface Shield queue is grouped: one job = one tank on one day, area asked once and credited once.	Store Keeper, HOD	§4.9a.2
What's new notices after an update, and tutorial freshness warnings for admins.	Everyone; Admin	§3.8, §7.14
The new 3D login (a still crest on phones and older computers).	Everyone	§3.1
Stock vs the Excel workbook: the Stock page marks materials whose stock differs from the workbook's, and says why.	Store Keeper, HOD, Admin	§3.9
An entry saved while the signal dropped is sent once, however many times it is retried.	Store Keeper	§3.7

3.11.2 Phase 15 - Practice, Garnet and the store keeper's note (28-30 September)

Change	Who	Read
Practice is violet with a pulsing badge, and refuses to start on an out-of-date database (no more half-working pages).	Everyone	§26.2, §17.9
Garnet is surface preparation: it credits no area of its own; the HOD answers Old or New surface per job.	Store Keeper, HOD	§4.9a.7
The store keeper's note fills the job card: Floor - 13.37 SQM Done fills the area, the part and the remark. Two notes on one tank and day are two jobs. The remark goes to the	Store Keeper, HOD	§4.9a.2

Change	Who	Read
HOD, the job history and the exports.		
The HOD adds and edits inventory items for their own site (Records -> Inventory -> New item / Edit). Deleting stays Admin-only.	HOD	§2.3.4, §2.3.8-2.3.9 (Adding an inventory item)
A user who belongs to one site no longer sees a Site box on the forms: their site is filled in and shown, not asked. Admins and multi-site users still choose.	Store Keeper, HOD	this section
The WBS No. field appears on Receive and Issue. If the HOD has not set up any WBS numbers yet it shows "None set up" and the entry still saves.	Store Keeper, HOD	§13.14, §16.5
Site and Category on a new item are matched to the spelling already in use (cncec -> CNCEC).	Admin, HOD	§2.3.8-2.3.9 (Adding an inventory item)

3.11.3 Phase 16 - lots, FEFO and the Lot Register workbook (1 October)

Change	Who	Read
The workbooks' Serial No. is read by the item: the batch for a Surface Shield, the roll for CHEMOLINE, the asset tag for equipment. Live now has real lots.	Everyone	§3.10.1
The Lot Register workbook (Rubber & Brick Materials - CNCEC.xlsx) adds each lot's MFD, expiry and rolls. It never changes stock.	Admin	§3.10.2
Lots & Expiry page: every lot, its expiry, and what is left of it.	Store Keeper, QC, HOD, Head of Qualities	§3.10.3
The Issue form's Lot field lists lots in FEFO order; an expired lot is listed last and never suggested. Choosing another lot asks for a reason (it never blocks).	Store Keeper	§3.10.3
The Receive form asks for the MFD; a blank expiry becomes MFD + shelf life.	Store Keeper	§3.10.3

Change	Who	Read
One evening expiry notice per site lists lots expiring within 30 days.	Store Keeper, HOD	§3.10.3
A return gives back to its lot, so a lot's balance is received - consumed - returned.	Everyone	§3.10.1
Practice has example lots: an expired one, a FEFO one, CHEMOLINE rolls, and a lot used but never received.	Everyone (training)	§3.10.4

3.12 The Hub Assistant answers some questions directly (Phase 17, October 2026)

The Hub Assistant (the round button at the bottom right of every page) now takes a quick look at your question before answering it - about half a second - and decides what kind of question it is. Four things can happen:

You ask...	You get
to open a page - "open the lots page", "go to Reports"	An Open ... button for that page. Only pages your role can open are ever offered; a page you cannot open, or a name that fits several pages, gets an ordinary answer instead. It only opens pages - it never does anything for you.
for a video - "is there a video on staging a return?"	The Watch it button, straight to the moment in the training video that shows it. If no video covers it, you get an ordinary written answer.
for numbers or records - "show me the receipts from last week", "which lots expire in the next 30 days?"	A small table in the chat. This is the same data as Reports -> AI -> Ask your data, and only the roles that already have that card get it: HOD, Logistics, Auditor and Admin. An HOD sees their own site only. Everyone else gets the usual written answer from the manual.
anything else - how to do something, what a word means, who can do what	A written answer from your section of the manual, exactly as before.

A question that tries to trick the assistant - telling it to ignore its rules, asking for its hidden instructions, typing database commands - is refused with the usual sentence: "I can only answer questions about your section of the manual." Ordinary work sentences that happen to use the same words ("drop the damaged drums at bay 3", "ignore the damaged drum and issue the rest") are answered normally.

If the quick check is not available (the AI service is restarting, or an admin has switched it off), the assistant simply answers as it always did. You lose nothing; only the buttons and tables do not appear.

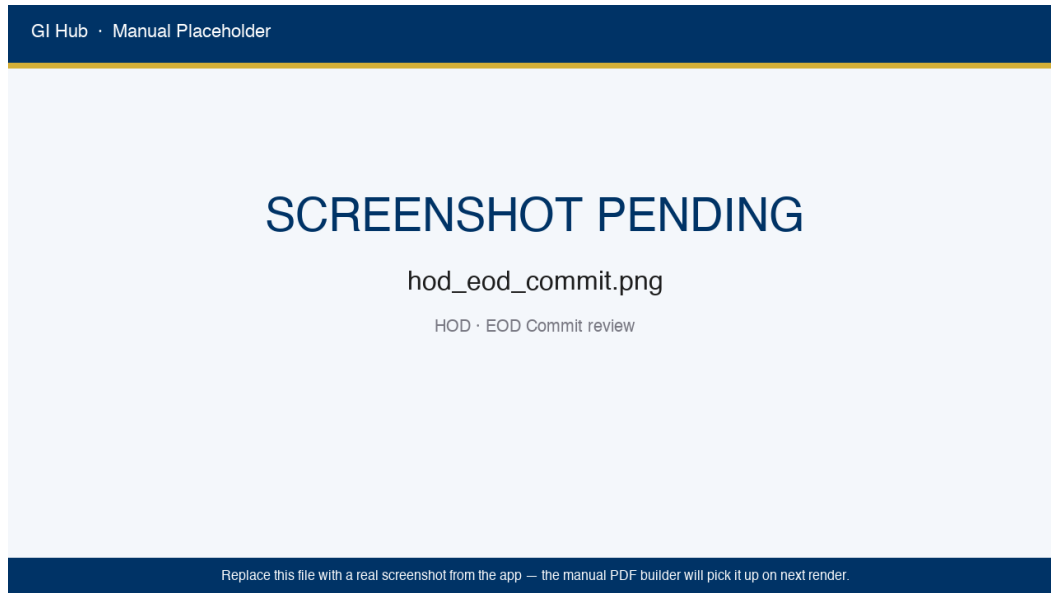
For admins: the quick check can be switched off without a restart - Admin -> Console -> Settings, set `ai_router_enabled` to 0 (and back to 1). The AI health line shows whether it is on and ready.

Two smaller changes arrived with it:

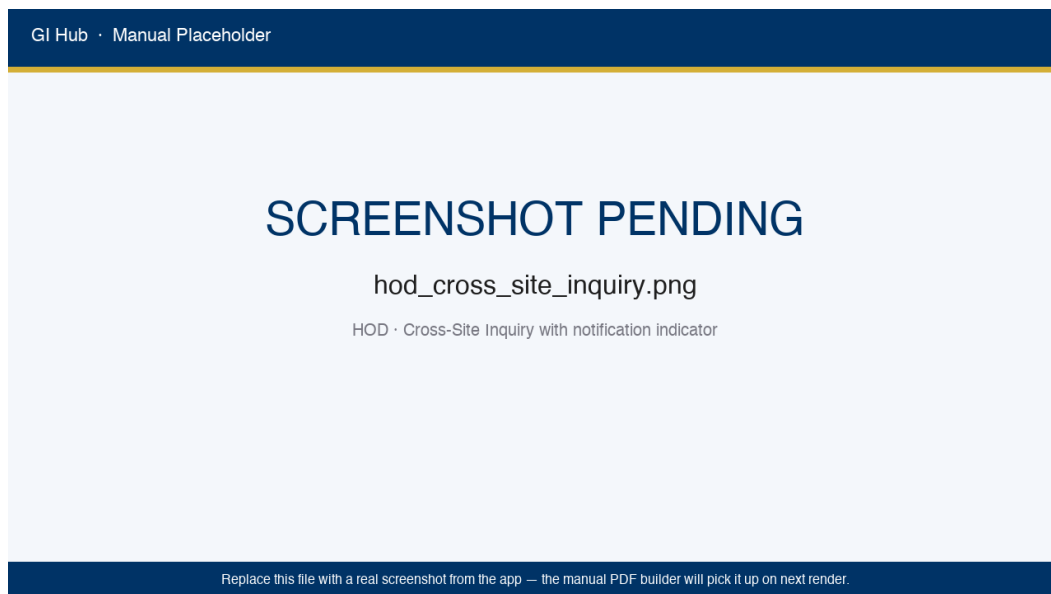
Change	Who	Read
The Head of Qualities no longer sees Reports or the Records ledgers (Receipts, Consumption, Returns, Lots, Purchase Requests) in the menu. They never worked for this role - every one showed an error - and the role's pages are Quality Oversight and Lots & Expiry.	Head of Qualities	§2.2, §2.3.5
The Watch it button appears less often by mistake. A question that merely says "video" or "tutorial", or shares only a word like "many" or "into" with a video, no longer gets a link to the wrong moment.	Everyone	§24.2.3

6. HOD (Head of Department) Manual

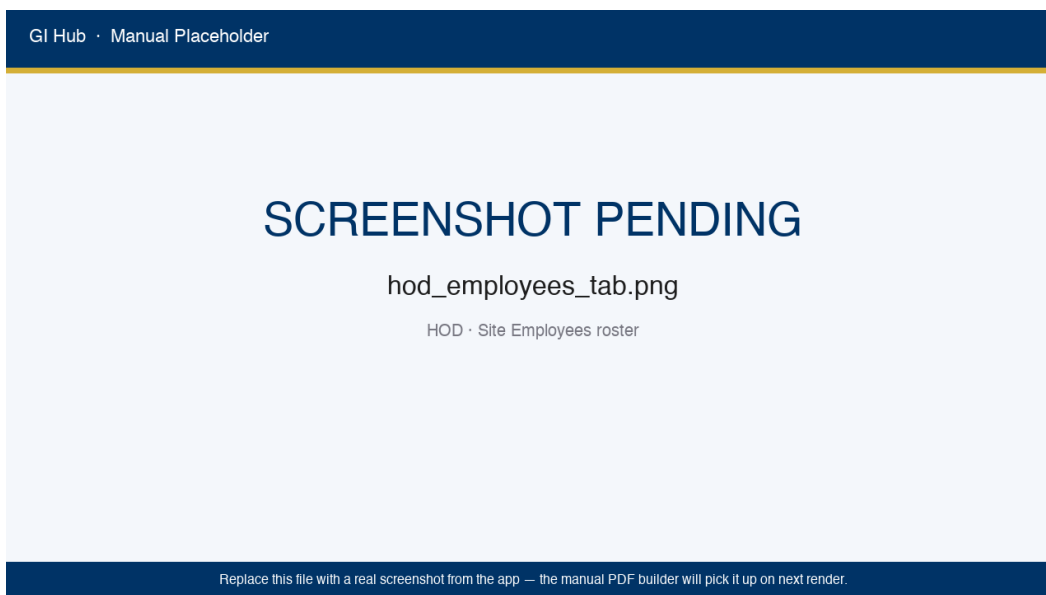
The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has 16 tabs as of v3.0 - covering EOD, Cross-Site, Burn Rate, Receipts, Returns, Adjustments, PRs, Shelf-Life, Notifications, My Requests, Site Config, the new Employees tab (Phase 7A), DOC, QR Approval, DN Approvals, In-Transit.



HOD Portal - End-of-Day Commit review with negative-stock guard



HOD Portal - Cross-Site Inquiry with the Phase 7C "you are viewing" notification banner



HOD Portal - Site Employees roster with add/edit form (Phase 7A)

6.1 Pages visible

- Live Dashboard (see §5.2)
- Entry Log (see §4 - HODs can also stage entries)
- HOD Portal - detailed below
- Reports (see §8)

6.2 HOD Portal overview

6.2.1 Page header

Reads "HOD Management Portal" - purple subtitle + today's date.

- Page title: " HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	Total value of stock held at your site	Delta: SAR consumed in the last 30 days - site-scoped
Below minimum (site)	Count of items at your site below their reorder threshold	Green/amber/red
Expiring / expired	Count of short-dated and expired items at your site	Green/amber/red
Pending receipts to approve	Receipts at your site waiting on your approval	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit
2. Cross-Site
3. Burn Rate
4. Pending Receipts
5. Adjustments
6. Purchase Requests
7. Receive Material
8. [!] Shelf-Life
9. Notifications
10. [OK] My Requests

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Approves every pending row in one action. Disabled when there are none.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing every column from the staging row (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects Lot_Number and FEFO_Override columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row
[X] button	Reject this single row

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click Commit EOD to Master:

Step 1 - pre-flight: negative-stock guard

Before committing, the system checks that nothing would push stock below zero. If anything would:

Element	Purpose
Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume - Deficit
Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort
Confirm Commit button (red)	Moves every approved row out of staging and into the permanent consumption ledger, refreshes the stock figures, and - if the commit leaves anything below its minimum - queues a post-EOD low-stock alert to the HOD.

A confirmation appears once the commit succeeds.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours

Element	Purpose
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	Whichever is smaller - the quantity you asked for, or the quantity actually available at the target site
Add to List button	Adds to the cart

6.4.3 Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes
Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site HOD via separate WhatsApp + audit log
Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests addressed to their site from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Marks the request approved and writes an audit entry
Rejection action	Marks the request rejected, also audited

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)
- Per-row line: material name (truncated) - gradient bar (color reflects burn intensity proportional to max) - monthly figure with UOM - days-remaining badge (red if <=7d, amber if <=30d, green if >30d)

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed forecast table

Columns: SAP_Code - Equipment_Description - UOM - Current_Stock - Daily_Burn_Rate - Days_Remaining

Sortable + exportable.

6.6 HOD Portal -> Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: " N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Approves every pending receipt at your site in one action. For each one, the system:

1. Works through every pending receipt in turn
2. Records the delivery (creating a lot record where an expiry date was entered)
3. Posts it to the permanent receipts ledger
4. Auto-closes the PR if it is now fully fulfilled
5. Removes the staged row
6. Refreshes the stock figures, writes an audit entry, and confirms with a success message

6.6.3 Receipt table

Columns: Date - SAP - Equipment_Description - UOM - Quantity - Supplier - PR_Number - Site_ID - Status pill

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier

Element	Purpose
[X] Reject button	Marks the row rejected rather than deleting it, so the history is preserved, and records your reason in the audit log

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
Pending	Count of pending adjustments at your site
Shortfalls	Variance < 0 count
Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - Variance (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Approves the adjustment and, in the same step, posts a matching correcting entry to the ledger - a shortfall is written as a consumption entry marked "stock adjustment"; a surplus is written as a receipt from supplier "stock adjustment". The adjustment keeps a reference to the ledger entry it created, so the two can always be tied together in an audit.
[X] button	Rejects the adjustment. The row is kept, marked rejected, so the audit trail stays complete.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - Status pill - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Creates the PR as an open request in draft state

6.8.2 Upload PR PDF (details read automatically from the file) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	Reads the PR number and the material codes (anything in the GI-XXXXXXX format) out of the document, matches them against the material codes in the inventory master, and creates the PR lines
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - Workflow pill

Qty Pending = Requested Qty - everything already received against that PR number.

6.8.4a Submit PR(s) to Logistics Portal (NEW in v3.0)

A new expander sits between the PR list and the email/PDF block. It opens the procurement-chain path - handing off your PR to the in-app Logistics queue instead of (or in addition to) the legacy email path.

Element	Purpose
Multi-select	All open PRs at your site that haven't yet been submitted to Logistics
Submit Selected to Logistics button	The PR is marked as submitted to Logistics and appears

Element	Purpose
	in Logistics Portal -> Incoming PRs. A "PR submitted to Logistics" notification fires to the Logistics inbox, and to WhatsApp if that event is enabled.

The legacy email path (§6.8.4) still works - your team can use it for direct-to-vendor relationships that don't go through central Logistics. Both paths coexist. The email path is marked for future deprecation once procurement chain adoption reaches the agreed threshold.

6.8.4 Notify Logistics section

Shown whenever at least one PR is still open:

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table. Windows: opens Outlook via COM with HTMLBody. Fallback: mailto: with monospace plain-text table.
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts

Field	Type	Purpose
		(Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	Posts straight to the receipts ledger, and creates a lot record if an expiry date is set. Auto-closes the PR if it is now fulfilled, and refreshes the stock figures.	

6.9.3 Receipt History (last 50)

Below the form, a table showing recent receipts for this site: entry no. - Date - SAP - Material - Quantity - Supplier - PR_Number - Expiry_Date.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
Critical (<=30d)	Days_left 0-30
Warning (<=90d)	Days_left 31-90

6.10.2 Action-required banner

When expired_n > 0: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - Days Left - Status pill

6.10.4 Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason expired_disposal.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Adds the message to the outbound queue; the background sending service picks it up from there. The send is audit-logged as a manual WhatsApp.

6.11.2 Alert Thresholds card

Three sliders. Each is remembered until it is changed again:

Slider	Range	Default	What it controls
Low stock alert (days of supply)	1-60	5	How few days of cover before an item counts as low
Burn-rate warning (days remaining)	1-60	7	How close to running out before the burn banner appears
Expiry warning (days before)	1-120	30	How far ahead of expiry an item starts being flagged

Save Thresholds button - saves the values and writes an audit entry recording the change.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - Status pill - Created

6.12.2 Mark Incoming Transfers as Received

Shown whenever at least one of your requests has been approved:

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Marks the request fulfilled and refreshes the stock figures. This is where you confirm the transfer physically arrived.

6.15 HOD Portal -> DN Approvals (NEW in v3.0)

This is your approval queue for Delivery Notes inbound to your site. The Warehouse has prepared them, Logistics has approved the delivery date - now you confirm the content (what's actually arriving, in what qty).

6.15.1 Empty state

If nothing's pending: an empty-state card reads "No DN's awaiting your approval - They appear here after Logistics signs off the delivery date." No action needed.

6.15.2 Per-DN cards

Each pending DN renders as its own bordered container with two columns:

Left column:

- Header line: DN <number> - PO <number> - From Warehouse <id>
- Subline: DN Date - Vehicle No - Driver Name
- View lines expander: Read-only preview of every line item (Material_Code, Description, Qty, UOM, rl_bl_family, Lot_Number, Expiry_Date, Remarks)

Right column (Decide popover):

- Notes textbox (required if rejecting)
- [OK] Approve button (primary) - flips DN to pending_sk and mirrors lines into pending_receipts so SK sees them
- [X] Reject button - flips DN to rejected with your reason; Warehouse gets pinged to redo

6.15.3 When to approve

- DN qty matches what your site needs (matches the PR line you originally raised)
- RL/BL family is correct
- Lot + expiry are acceptable (not over-expiry)
- The originating PO is the right one (not a misroute)

6.15.4 When to reject

- Wrong qty (e.g. site asked for 100, DN says 200 - too much exposure)
- Wrong material (the Material_Code doesn't match the PR you raised)
- Lot has insufficient remaining shelf life
- Vehicle / driver details suggest a routing problem

Always include a clear rejection reason. The Warehouse user sees the reason verbatim in their bell inbox and on the bounced DN. Sloppy reasons cost a rebuild cycle.

6.16 HOD Portal -> In-Transit (NEW in v3.0)

A read-only window onto the procurement chain for your site. Use this tab when a user asks "when is X arriving?" or "why didn't Y arrive yesterday?". Three sub-tabs.

6.16.1 Sub-tab: Active in-transit

KPI strip at the top with counts per pipeline state:

- At Logistics - DN drafted by Warehouse, waiting on Logistics date approval
- Logistics approved - date confirmed, waiting on YOU
- Awaiting my approval - same as Logistics approved (highlighted gold)
- Pending SK receipt - you've approved, SK has it in their tab

Below the KPI strip, each in-transit DN renders as a card:

Element	Purpose
DN header	DN <number> (gold) + RL/BL chip (orange for RL, purple for BL)
Subline	PO <number> - Warehouse <id> - ETA <date> - <line count> line(s), <total qty> units
Status pill (right)	Colour-coded pipeline state
View lines expander	Read-only line preview
Request reschedule popover	Date picker (defaults to ETA + 3 days, min=today) + reason textarea + Submit button. Submits to Logistics.

The reschedule UI is deliberately frictionless:

- Date is pre-filled so a quick "+3 days" submission is one click
- min_value=today prevents accidentally picking yesterday
- Same-date submission warns instead of wasting a round-trip
- Caption " Goes to Logistics" makes the destination unambiguous

6.16.2 Sub-tab: My reschedule requests

KPI strip: Pending / Approved / Rejected counts.

Custom table showing your full reschedule history for THIS site:

Column	Source
PO No.	the linked PO
DN No.	the linked DN (if any)
From	current_date when you raised the request
Requested	the new date you asked for (gold, monospace, with arrow)
Reason	your justification (tooltip-truncated)
Status	pill - pending / approved / rejected
Decided by	Logistics user who handled it
Notes	decision notes from Logistics

6.16.3 Sub-tab: Force-closures affecting me

Read-only audit table showing every PR / PO / line that Logistics force-closed, scoped to your site. Where a closure record itself carries no site, the system traces it back through its parent PR or PO, so nothing goes missing from your list.

Column	Source
Type	"PR closed" / "PO closed" / "Line closed" badge
Target	The closed ref (PR number / PO number / line id)
PR No.	Linked PR (if applicable)
PO No.	Linked PO (if applicable)
Reason	Logistics' force-close reason
Closed by	Logistics user who closed it
When	Timestamp

50 most recent shown.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit
2. Review the 4-card stat strip
3. Sort/filter using the filter pills
4. For each row: read the full detail card (line 1 = essentials, line 2 = all fields)
5. Click [OK] to approve or [X] to reject individual rows
6. Optionally click [OK] Approve All Pending for a clean batch
7. Click Commit EOD to Master
8. If pre-flight blocks: read the violation table, close, fix the offending rows (reduce qty, drop, or receive stock first), then re-open
9. If pre-flight passes: type COMMIT exactly, click Confirm Commit
10. Watch the balloons.

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments
2. Read the pending table - variance and reason are color-coded
3. For each adjustment, decide: did the Store Keeper count accurately?
4. [OK] Approve -> system posts a synthetic ledger row (receipt or consumption), live stock updates immediately
5. [X] Reject -> audit-logged; Store Keeper sees a status update

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site
2. Pick target site -> pick material -> see live availability
3. Add to cart, repeat
4. If cart > 5 items: an automatic WhatsApp escalation to the target HOD will fire on submit. Document why in Justification.
5. Click Submit All Requests to Admin
6. Watch the [OK] My Requests tab - when status flips to "approved", you can mark it as fulfilled once received

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests
2. Review the request table
3. [OK] Approve -> physical handoff begins
4. [X] Reject -> use when the request is unjustified or your stock is too tight

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests
2. Create New PR
3. Type PR Number, pick Material (auto-fills SAP, MatCode, UOM), Qty, Supplier, Est. Cost SAR, Notes
4. Click Create PR Draft

Option B: PDF upload

1. Upload PR PDF
2. Drop the PDF in
3. Click Process Upload
4. System auto-extracts PR number + items, creates pr_master rows

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests
2. Select open PR in the email dropdown
3. Draft Outlook Email -> Mail/Outlook opens with HTML table pre-filled
4. Review/send
5. Also: Download PR PDF for your records

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds - the stock figures refresh on that cycle. Refresh the page. If it still doesn't, check the audit log for the adjustment-approved entry to confirm the correcting ledger entry was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in receipts table, not pending_receipts.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is mailto: (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format GI-XXXXXXX. Items not in your inventory's Material_Code field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> Audit Logs by the FEFO override action. Every consumption entry that used an override also carries the override reason on the row itself.

6.17 SME -> QR Excel - when the paper and the book both record a drum

A supervisor files a QR form for a job. The store keeper also logs the same drums in the Excel Consumption Log. GI Hub now makes sure they are counted once: for each day - equipment - material the stock holds the larger of the two figures, never both added together.

What you see	What it means	What to do
Matched	The paper and the book agree.	Nothing.
Excel shows more	The book has extra drums the form did not list. The extra is added once and attached to the same job. No second area is credited.	Nothing, unless the extra is wrong.

What you see	What it means	What to do
Awaiting Excel	The form is filed; the workbook has not caught up yet.	Nothing. It matches when the book is synced.
Conflict - paper claims more	The form says more than the book. The stock holds the form's figure. You are notified.	Check the job. If the form is wrong, correct it on the execution entry. If the book is wrong, fix the workbook and re-sync.
Counted twice - fix at source	Both the form and an edited book line cover the same drums, above both figures.	Correct the entry or the workbook. Nothing is rewritten automatically.
Possible duplicate (+/-1 day)	An Excel draw of the same material on the same equipment one day from a QR form. It was not merged; only the exact day merges.	Check whether it is the same drum. If it is, remove or correct the workbook line.

Which order things happen in does not matter. If the book already had the drums when the form arrives, the form adopts them and only adds what is missing. If the form came first, the book's lines are matched against it. Equipment names are matched through the Tank Aliases screen, so a different spelling of the same tank still lands on the same vessel.

Acknowledge records that you looked at a conflict or a possible duplicate. It changes no quantity.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

The header reads "Reports & Analytics". Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply remaining per item	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first,	Answering "what is on our shelves"

Report	What it shows	Typical use
	with the top-ten share	worth?"
Shelf-Life / Expiry	Lots grouped into expired, critical and warning	The weekly walk-round to pull short-dated stock
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first, and where they did not	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Builds the report for the type, date window and site you chose. The page shows a loading shimmer while it works.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	The report type and site, above the date range covered
v Download button	Saves the report in the format you picked - PDF, Excel or CSV
Summary cards	The headline figures for the report, filled in automatically
Bar chart	Shown for the Burn Rate and Daily Consumption reports
Preview table	The rows themselves, with status colours

8.2.5 A note about apostrophes in downloaded files

Sometimes a downloaded Excel or CSV file shows an apostrophe in front of a value you typed - for example '=A1 where you typed =A1. Nothing is wrong, and nothing has been lost.

Spreadsheet programs treat anything starting with an equals sign, a plus, a minus or an at-sign as a calculation to run, not as words to display. If that text came from something a person typed into the system, opening the file could make the spreadsheet run instructions nobody intended. The apostrophe tells the spreadsheet "this is text, show it exactly as it is".

- It only ever appears in front of text that begins with one of those symbols.
- Numbers are never affected. Quantities, prices and totals stay numbers, and every total in the file still adds up correctly.
- Ordinary words, notes and material names are untouched.
- If you need the plain text back, delete the leading apostrophe in the cell.

You will see this most often on notes and remarks fields, since those are the places people type freely.

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st 06:00
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label
- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

Available when AI is switched on and the local AI service is running:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 standard checks and rewrites the AI commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	How sure the AI is of this finding
Right-side metric callout	Headline number with sub-label
Body paragraph	The AI's explanation, written from the figures the check returned
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 standard checks:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If the local AI service is not running, the tab explains what an Admin needs to start.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
 - Re-email
 - Delete (audit-logged)
-

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (the unique identifier), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current Stock} = \text{Total Received} - \text{Total Consumed} - \text{Total Returned}$. It is never stored as a number anyone can type over; it is recalculated from the ledger every time it is shown.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	Access tokens for the offline-capable mobile app
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered -> closed / force_closed / cancelled
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered / returned / closed / force_closed
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed / cancelled
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned / cancelled
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled

Table	What it stores	State machine
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	Unread -> read, stamped with the time it was read
delivery_reminders_sent	Record of which T-2 / T-1 / T-0 reminders have gone out	Write-once - the same reminder can never be recorded twice

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

26. Practice Mode - learn the system without touching Live

Practice is a separate copy of GI Hub for training. It has every page, every role and every workflow Live has, but it runs on its own made-up data in its own database. Nothing you do in Practice can reach Live: not a receipt, not an approval, not a message.

Use it to learn a job before doing it for real, to rehearse something you do rarely (a return, a stock count, a PR), or to try something out without worrying about the result.

26.1 Switching between Live and Practice

On the sign-in screen, above the username box, there is a Live | Practice switch.

- Live is selected unless you choose otherwise. It is the real system.
- Choose Practice and the page reloads, ready to sign in to Practice.
- Choose Live again to go back.

Things to know about switching:

- Switching signs you out. You sign in again in the environment you picked. This is deliberate: a browser is in one environment at a time.
- Your Live password does not work in Practice, and a Practice password does not work in Live. They are separate systems with separate accounts (§26.3).
- The installed apps (Windows, macOS, Android, iOS) have the same switch.

26.2 How you know you are in Practice

The page tells you in six ways, and none of them can be dismissed:

Where	What you see
Colours	Practice is violet: the sidebar, the header, the page background and the main buttons. Live is navy and gold.
Top left, under the GI Hub logo	A pulsing amber PRACTICE badge, on every page. On a phone it sits beside the menu button. On the sign-in page it appears at the top left as soon as you choose Practice. With Reduce motion on, it stays still.
Top of every page	An amber bar: PRACTICE - practice data, nothing here is real.
Page header	An amber PRACTICE tag beside the title, and an amber line under the header that stays when you scroll
Browser tab	The tab title starts with PRACTICE -

Where	What you see
Anything you print	A large diagonal PRACTICE - NOT REAL DATA watermark

These come from the server you are connected to, not from the switch. The server reports which environment it is, and the page shows what it says.

[!] If you see a red bar saying "Environment mismatch", the switch and the server disagree. Everything you try to save will be refused until that is fixed. Sign out, pick the environment again, and tell your administrator if the red bar comes back.

26.3 The practice accounts

Practice has one shared account per role. Your trainer gives you the password for these accounts.

Role	Practice account
Store Keeper	practice.storekeeper
Supervisor	practice.supervisor
Head of Department	practice.hod
Quality Control	practice.qc
Head of Qualities	practice.qchod
Warehouse	practice.warehouse
Logistics	practice.logistics
Auditor (view-only)	practice.auditor
Admin	practice.admin (separate password, held by your trainer or administrator)

The site-based accounts work at site CNCEC, and the Warehouse account works at warehouse WH-01.

Everyone in a class uses the same accounts, so you will see each other's practice entries. That is normal.

26.4 What is in Practice

The data is made up but realistic. The categories, units of measure, lining systems, work types and tank numbers are real. The material names, SAP codes, people, vendors, quantities, prices and dates are all invented. Nothing in Practice is a real person or a real stock figure.

It is the same data the training videos use. If a video shows Tutorial Demo Gasket Set (SAP 899001), you can find it in Practice.

Work is already waiting so every role has something to do: receipts, issues and a return waiting for HOD approval, a draft purchase request, and one already with Logistics.

Every new Live feature comes with a Practice example (rule 17g), so it can be tried before it is used for

real. The current ones:

Feature	Try it on
The store keeper's note filling the job card (§4.9a.2)	One tank and day with two notes, Floor - 12.5 SQM Done and Sump Wall - 4.2 SQM Done: two jobs, each with its area and part filled in.
Lots, FEFO and expiry (§3.10.4)	PRACTICE PU PRIMER (899971): PR-OLD (expired), PR-SOON (the FEFO lot), PR-LATE.
CHEMOLINE rolls (§3.10.4)	PRACTICE CHEMOLINE (899973), batch 1026009999, three rolls.
Lots used but never received (§3.10.3)	Lot PR-TYPO.
The assistant's quick answers (§3.12)	As practice.hod, ask the Hub Assistant "which lots expire in the next 30 days?" - a table, from Practice's own lots. As practice.storekeeper, ask "open the lots page" - an Open Lots & Expiry button - and "is there a video on staging a return?". Nothing new had to be added: the quick check reads whatever database the Practice process holds (rule 17).

26.5 What works differently in Practice

Everything works exactly as in Live, except these four things. Each one is switched off on purpose:

In Practice...	Why
No WhatsApp message or email is ever sent. Notifications still appear in the bell.	A practice alert must never reach a real phone or inbox.
Photo reading (OCR) is switched off. The page explains this, and you can still paste text or type rows in.	The AI engine is shared with Live and takes minutes per page, so trainees must never queue ahead of real forms.
Training videos watched in Practice are not counted. You can watch as much as you like, but I have watched and understood this is refused.	Training records are kept in Live only. Sign in to Live to have a video counted.
Two-factor authentication is off, and cannot be turned on.	The accounts are shared, so one person's authenticator would lock everyone else out.

26.6 Offline entries stay where they were made

If you lose signal while saving an entry, GI Hub keeps it and sends it when the connection returns (§3.7). In Practice this still works, with one extra guarantee: an entry saved in Practice is only ever sent to Practice. If you switch to Live before it has been sent, it waits, and it is sent the next time this browser is back in Practice. It can never be sent to Live, even if you sign in to Live on the same device.

26.7 Resetting Practice (admins)

Over time Practice fills up with everyone's practice work. An admin can put it back to its starting state:

1. Sign in to Practice as practice.admin.
2. Open Admin Console -> Practice. The page shows when the data was built and when, and by whom, it was last reset.
3. Click Reset Practice data... and type RESET PRACTICE DATA to confirm.

The reset takes about a second. It removes everything anyone has done in Practice, and everyone signed in to Practice is signed out. The approval queues are filled up again. Live is not affected in any way. It is a different database, and the Practice server has no permission to open it.

The Practice tab and the reset only exist in Practice. In Live there is no such button, and the server has no such function.

How administrators set up and run Practice on a server is in §17.9.